

# CS F407: Goal-Based Warehouse Agent

AI Laboratory

## Problem and design

The environment is the supplied static warehouse grid. The state is a zero-based (*row, column*) position. The goal is to reach G from S without crossing a shelf. Actions are up, down, left, and right; each costs one. The agent stores the map, its position, goal, search frontier, discovered positions, and predecessor of each discovered position. It is goal-based because its action sequence is selected to reach an explicit destination. A reflex rule alone does not provide a planned route around obstacles.



BFS uses a FIFO queue and finds a shortest path under unit movement costs. Each reachable position is discovered once. For  $V$  traversable cells and  $E$  legal moves, time is  $O(V + E)$  and memory is  $O(V)$ . A larger warehouse increases the frontier and memory use. Manhattan A\* may reduce exploration; moving obstacles require replanning.

## Results

The supplied map has uneven row lengths. Bounds use each row's length, without modifying the map. Expansion counts exclude the goal.

| Test              | Found | Moves | Expanded |
|-------------------|-------|-------|----------|
| original          | Yes   | 20    | 58       |
| adjacent          | Yes   | 1     | 1        |
| blocked           | No    | –     | 1        |
| alternative paths | Yes   | 6     | 14       |

**Original route.** Asterisks mark its intermediate cells.

```
#####
#S***.#####G#
#.#####.##
#...##.....#
#.#####.###.#.###.##
#.....#.....#
#####
```

Every route was checked for endpoints, legal moves, obstacle avoidance, and reported length. The adjacent and blocked tests have known outcomes. The alternative-path case has a shortest route of six moves. Full coordinates are in `results.json`.

## LLM evaluation

The first executed implementation passed all four tests. BFS was selected because moves have equal cost and a shortest route is required. A precise prompt states the map, legal moves, goal, failure behaviour, and outputs. Future prompts should also specify coordinates and row-bound handling. These tests support this implementation; they do not cover every navigation environment.

**Reproduce.** Run `../venv/bin/python solution.py` here. Results: `results.json`; prompt and correction record: `prompts.txt`.